PUSHFORWARD ALONG THE ETALE SIDE OF AN ELEMENTARY DISTINGUISHED SQUARE PRESERVES PERFECTNESS WITH SUPPORT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3617-3625, immediately following the statement at 3609-3616.

Editorial note. The U-side vanishing consumes the support hypothesis via the restriction lemma; the V-side is a direct identity.

Allowed standard prerequisites (editorial).

- Restriction of the supported pushforward. Lemma lemma-pushforward-with-support-in-open of this chapter, cited at proof lines 3620-3622. Rj_*E restricted to V equals E.
- Restriction commutes with Rf-star. Lemma lemma-restrict-direct-image-open of this chapter, cited at proof lines 3623-3625. Identifies Rj_*E restricted to U, enabling the support reading.

Source reference. Original lemma 08GG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2135–2135. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08GE: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 275–275. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $(U \subset X, j : V \to X)$ be an elementary distinguished square of algebraic space over S. Let E be a perfect object of $D(\mathcal{O}_V)$ supported on $j^{-1}(T)$ where $T = |X| \setminus |U|$. Then Rj_*E is a perfect object of $D(\mathcal{O}_X)$.*

Proof. Being perfect is local on $X_{étale}$. Thus it suffices to check that Rj_*E is perfect when restricted to U and V. We have $Rj_*E|_V = E$ by Lemma 08GG which is perfect. We have $Rj_*E|_U = 0$ because $E|_{V \setminus j^{-1}(T)} = 0$ (use Lemma 08GE). □

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.